



Irkutsk National Research Technical University
Baikalsk school of BRICS

09.04.02 Information systems and technology
Program: Information technologies, networks and big data

Course: Data analysis

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Data Manipulation and Cleaning

1. Missing Data
 - Identifying and dealing with NA values
 - Imputation techniques
2. Outliers
 - Outlier Detection.

Download or copy from Module 2: Data Manipulation and Cleaning
the script file **Data_Manipulation_Process.R**
the data file **DataSet_No_Details.csv** to your computer.

Data Set



Name: "Hormonal and Biochemical Profile in Patients"

Source: Internal company database

Size: 1148 rows × 41 columns

Format: CSV

Columns:

record_id (int) Unique order identifier

outcome (int) Presence of a tumor

hormone1 (num) result of hormone 1 level assessment

...

hormone4 (num) result of hormone 4 level assessment

h_index_34 (num) index (hormonal ratio)

...

hormone14 (num) result of hormone 4 level assessment

lipids1 (num) result of lipid 1 level assessment

...

lipids5 (num) result of lipid 5 level assessment

carb_metabolism (num) result of carbohydrate metabolism assessment

lipid_pero1 (num) result of lipid peroxidation 1 level assessment

...

lipid_pero5 (num) result of lipid peroxidation 5 level assessment

antioxidant1 (num) result of antioxidant 1 level assessment

...

antioxidant5 (num) result of antioxidant 5 level assessment

factor_eth (int)) Ethnicity

factor_h (int) Increased hormones

factor_pcos (int) Diagnosis

factor_prl (int) Increased hormone

Potential Use for the Practical class 1:

Data Manipulation and Cleaning (Missing Data&Outliers)

17.03.2026

	DataSet_1	DataSet_No_Detail		
1	record_id	record_id		
2	fibroids	outcome		
3	tsh	hormone1		
4	prl	hormone2		
5	lh	hormone3		
6	fsh	hormone4		
7	lh_fsh	h_index_34		
8	testosteron	hormone5		
9	shbg	hormone6		
10	FAI	h_index_56		
11	p_17hp	hormone7		
12	dheas	hormone8	27	DvSv lipid_pero1
13	p4d22_24	hormone9	28	DK lipid_pero2
14	estadiol	hormone10_1	29	KD_CT lipid_pero3
15	estradiol_2022	hormone10_2	30	MDA lipid_pero4
	estradiol_generate	hormone10_generate	31	AOA lipid_pero5
16	d	ated	32	COD antioxidant1
17	_2OH	hormone11	33	GSH antioxidant2
18	_16OH	hormone12	34	GSSG antioxidant3
19	kortizol	hormone13	35	GSH_GSSG an_index_23
20	estrone	hormone14	36	vitE antioxidant4
21	triglycerides	lipids1	37	vitA antioxidant5
22	cholesterol	lipids2	38	ethnicity factor_eth
23	hdl	lipids3	39	H_androg factor_h
24	ldl	lipids4		
25	vidl	lipids5	40	Phenotype_PCOS factor_pcos
26	glucose	carb_metabolism	41	hyper_prl factor_prl

Notes for the final project



Dataset Description Template

1. General Information

- **Dataset Name:** (e.g., "Customer Purchases 2023")
- **Source:** Where the data was obtained (e.g., company database, Kaggle, API, web scraping).
- **Author/Owner:** (e.g., "Company X," "UCI Machine Learning Repository")
- **Creation/Update Date:** (e.g., "Last updated: March 2024")
- **License:** (e.g., "CC0: Public Domain," "Commercial use restricted")
- **Purpose:** Why the dataset was created (e.g., "To analyze customer behavior").

2. Dataset Structure

- **Format:** (CSV, JSON, SQL, Parquet, etc.)
- **Size:**
 - Number of records (rows)
 - Number of features (columns)

• **Variables/Columns Description (Tab.):**

Column Name	Data Type	Description	Possible Values	Missing Values?
-------------	-----------	-------------	-----------------	-----------------

3. Data Content & Semantics

- **Domain/Context:** (e.g., "E-commerce transactions")
- **Target Variable (if applicable):** (e.g., "price for regression analysis")
- **Time Coverage:** (e.g., "Jan 2020 – Dec 2023")
- **Geographical Coverage (if applicable):** (e.g., "US customers only")

4. Data Quality

- **Missing Data:** Percentage and columns affected.
- **Duplicates:** Are there duplicate entries?
- **Outliers/Anomalies:** Unusual values (e.g., negative prices).
- **Data Balance:** For classification, check class distribution.

5. Usage Examples

- **Possible Analyses:** (e.g., "Customer segmentation, sales forecasting")
- **Machine Learning Tasks:** (e.g., "Classification: Predict product_category")

6. Additional Notes

- **Collection Method:** (e.g., "Web scraping, manual entry")
- **Preprocessing Done:** (e.g., "Normalized prices, removed duplicates")
- **Limitations:** (e.g., "Only includes US data, no refund records")

Missing Data



Strategy:

Identify Missing Data:

Where and how data is missing in the dataset  the nature of the missing
hypothesize about potential reasons for its absence

Quantifying Missing Data:

Determine the extent of missing data, both at a *variable level* and *observation level*

Visualizing Missing Data Patterns: Visual patterns of missing data sometimes can reveal insights, indicating whether missing data is random or systematic.

Analyzing the Impact of Missing Data:

Assess how missing values in one variable might relate to another, aiding in understanding if data is Missing Completely At Random (MCAR), Missing At Random (MAR), or Missing Not At Random (MNAR).

Handle Missing Data

Missing Data



up to 15% - can be restored safely,
15-30% - with caution,
>30% - it is often better to exclude or use special methods.

Key factors for decision making:

Missing type:

MCAR (*missing completely at random*): easier to impute

MAR (*missing at random*) / MNAR (*missing not at random*) : require complex methods

Variable importance:

Key predictors can be recovered even at 20-30%

Minor variables with >15% are sometimes better excluded

Sample size:

In large datasets (>10,000 observations) recovery can be attempted even at 20-25%

In small samples (>30%) often not recoverable

Analysis method:

For descriptive statistics - stricter restrictions

For ML models, higher percentages are sometimes acceptable

Required Software for Working with R



Basic Software

R language

Download from the official website: cran.r-project.org

(choose the version for your operating system: Windows, macOS, or Linux)

RStudio IDE

Download the free RStudio Desktop version: posit.co/download/rstudio-desktop/

RStudio provides a user-friendly interface with code completion, plot viewer, project management, and more.

Recommended Additional Components



Recommended Additional Components

Essential R packages

For practical class 1:

skimr - *skim()* summary with histograms for numeric variables

visdat - *vis_miss()* visualizes NA patterns

naniar - *gg_miss_var(MD_df)* barplot of missingness per variable

After installing R, open RStudio and run the following command in the console:

```
install.packages("skimr")
```

....

List of Commands Used in the Script



1. Working Directory & File Paths

getwd() – Returns the absolute path of the current working directory.

read.csv() – Reads a comma-separated values (CSV) file and creates a data frame.

2. Data Inspection

str() – Displays the internal structure of an R object (data types, first values).

skim() – (from skimr package) Provides a comprehensive summary of a data frame, including histograms for numeric variables and missing value counts.

summary() – Generates descriptive statistics (min, max, mean, quartiles) for each column.

3. Package Management

install.packages() – Downloads and installs R packages from CRAN.

library() – Loads an installed package into the current R session.

List of Commands Used in the Script



4. Data Manipulation (dplyr)

select() – Selects (or drops) columns from a data frame.

any_of() – A helper for select() that safely selects columns that exist (ignores missing names).

%>% (pipe) – Passes the result of the left-hand side as the first argument to the function on the right-hand side, enabling readable sequential operations.

5. Exporting Data

write.csv() – Writes a data frame to a CSV file.

List of Commands Used in the Script



6. Missing Value Analysis

is.na() – Returns a logical vector/matrix indicating which elements are missing (NA).

sum(is.na()) – Counts total missing values in an object.

colSums(is.na()) – Counts missing values per column.

colMeans(is.na()) – Computes the proportion (mean) of missing values per column.

data.frame() – Creates a data frame from given vectors or lists (used here to display missing-value statistics).

List of Commands Used in the Script



7. Visualizing Missing Data

vis_miss() – (from visdat package) Visualizes the pattern of missing values across a data frame.

gg_miss_var() – (from naniar package) Creates a bar plot showing the number of missing values in each variable.

8. Multiple Imputation with MICE

mice() – (from mice package) Performs multiple imputation using chained equations. Key arguments:

- m – number of imputed datasets.

- method – imputation method (e.g., "pmm" for predictive mean matching, "rf" for random forest).

complete() – Extracts one of the completed datasets from a mids object produced by mice()

List of Commands Used in the Script



9. Data Visualization (ggplot2)

ggplot() – Initializes a ggplot object; specifies the data and aesthetic mappings (aes).

geom_density() – Adds a kernel density plot layer.

geom_boxplot() – Adds a boxplot layer.

labs() – Sets labels for the plot (title, axis titles).

scale_x_continuous() – Modifies the x-axis scale (e.g., setting limits).

facet_wrap() – Creates a multi-panel plot by splitting data by a categorical variable.

theme_minimal() – Applies a minimal theme to the plot.

List of Commands Used in the Script



10. Data Reshaping (tidyr)

pivot_longer() – Collapses multiple columns into key-value pairs (long format), often used to prepare data for grouped plotting.

11. Other Helpful Commands

c() – Combines values into a vector (used to define column names to remove).

names() – Retrieves or sets the column names of a data frame.

! – Logical NOT operator (used with **%in%** to exclude elements).

%in% – Checks which elements of one vector belong to another.

<- (assignment) – Assigns a value to a variable.

Missing Data



```
> str(df)
'data.frame':   1148 obs. of  41 variables:
 $ record_id    : int  1 2 3 4 6 8 9 10 12 13 ...
 $ outcome      : int  0 0 0 0 0 1 0 1 1 0 ...
 $ hormone1     : num  0.5 1.9 1.2 3.8 1.6 1.6 0.42 2.7 1 1.8 ...
 $ hormone2     : int  314 196 311 409 418 309 415 450 198 569 ...
 $ hormone3     : num  9.9 6.8 1.2 3.2 3.8 2.7 5.9 7 9 20.6 ...
 $ hormone4     : num  4.2 6.8 3.2 2.7 7.2 1.6 9.1 4.5 5.2 8.3 ...
 $ h_index_34   : num  2.357 1 0.375 1.185 0.528 ...
 $ hormone5     : num  322.7 190.9 153.7 225.7 99.9 ...
 $ hormone6     : num  166.1 41 140.8 93.4 24.7 ...
 $ h_index_56   : num  0.674 1.615 0.378 0.838 1.402 ...
 $ hormone7     : num  5.7 6.8 5.4 12.7 1.6 9.9 7.5 4 4.2 1.7 ...
 $ hormone8     : num  129 260 81.1 148 121 120 135 199 149 238 ...
 $ hormone9     : num  NA NA NA NA NA NA NA NA NA NA ...
 $ hormone10_1  : num  NA NA NA NA NA NA NA NA NA NA ...
 $ hormone10_2  : num  0.922 NA 0.405 NA 0.199 0.499 0.815 0.798 0.615 0.506 ...
 $ hormone10_generated: num  0.922 0.166 0.405 0.441 0.199 ...
 $ hormone11    : num  NA NA NA NA NA NA NA NA NA NA ...
 $ hormone12    : num  NA NA NA NA NA NA NA NA NA NA ...
 $ hormone13    : num  NA NA NA NA NA NA NA NA NA NA ...
 $ hormone14    : num  NA 52.6 74.2 NA NA NA NA NA NA ...
 $ lipids1      : num  0.4 0.54 0.45 0.38 0.48 0.34 0.49 0.62 1.1 0.51 ...
 $ lipids2      : num  5.05 3.9 3.5 4.84 3.96 3.85 3.93 4.3 3.89 3.72 ...
 $ lipids3      : num  1.23 1.67 1.29 1.32 1.85 1.42 1.26 1.3 1.11 0.78 ...
 $ lipids4      : num  3.64 1.98 2.01 3.35 1.89 2.28 2.45 2.72 2.28 NA ...
```

Missing Data



```
> skim(df)
```

```
— Data Summary —
```

	Values
Name	df
Number of rows	1148
Number of columns	41
Column type frequency:	
numeric	41
Group variables	None

Missing Data

— Variable type: numeric —

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
1 record_id	0	1	1306.	746.	1	655	1258	1946.	2694	
2 outcome	1	0.999	0.139	0.347	0	0	0	0	1	
3 hormone1	14	0.988	1.81	1.51	0.05	1.1	1.5	2.1	26.1	
4 hormone2	16	0.986	382.	259.	31	228.	320	458	2798	
5 hormone3	19	0.983	9.12	11.7	0	3.5	5.7	9.4	118.	
6 hormone4	17	0.985	7.80	11.4	0.1	3.8	5.5	7.4	131.	
7 h_index_34	19	0.983	1.34	1.13	0	0.674	1.02	1.67	18	
8 hormone5	18	0.984	327.	323.	0	178.	270.	378.	5687.	
9 hormone6	16	0.986	78.7	53.5	0.76	40.7	65	101.	375.	
10 h_index_56	20	0.983	2.25	4.48	0	0.792	1.39	2.52	120.	
11 hormone7	16	0.986	5.64	3.92	0	2.68	5.1	7.6	47.1	
12 hormone8	19	0.983	176.	85.8	16.3	116	162	218	558	
13 hormone9	1009	0.121	23.8	29.1	0.1	5.75	10.9	30.4	172.	
14 hormone10_1	1078	0.0610	0.368	0.186	0.178	0.250	0.280	0.437	1	
15 hormone10_2	632	0.449	0.984	11.5	0.077	0.278	0.44	0.613	261	
16 hormone10_generated	393	0.658	0.843	9.49	0.077	0.284	0.432	0.632	261	
17 hormone11	987	0.140	12.3	4.68	1.1	9.7	13.1	16	19.2	
18 hormone12	987	0.140	13.1	3.97	0.5	10.7	14.2	16.4	19	
19 hormone13	954	0.169	441.	320.	68.9	196.	374.	535	1901	
20 hormone14	931	0.189	52.1	57.0	3.2	17.2	24.2	71.9	332	
36 antioxidant4	42	0.963	7.86	3.17	0	5.45	7.34	9.68	25	
37 antioxidant5	40	0.965	0.690	0.289	0.23	0.48	0.62	0.84	2.04	
38 factor_eth	1	0.999	1.46	0.659	1	1	1	2	3	
39 factor_h	14	0.988	0.194	0.396	0	0	0	0	1	
40 factor_pcos	983	0.144	2.6	1.20	1	1	3	4	4	
41 factor_prl	15	0.987	0.0671	0.250	0	0	0	0	1	

Missing Data



data set preparation

Remove
columns containing
a **high percentage of**
missing data

Separate Factor
Variables

Result

MD_df

factor_data.csv

```
> str(MD_df)
'data.frame':   1148 obs. of  31 variables:
 $ record_id      : int  1 2 3 4 6 8 9 10 12 13 ...
 $ hormone1       : num  0.5 1.9 1.2 3.8 1.6 1.6 0.42 2.7 1 1.8 ...
 $ hormone2       : int  314 196 311 409 418 309 415 450 198 569 ...
 $ hormone3       : num  9.9 6.8 1.2 3.2 3.8 2.7 5.9 7 9 20.6 ...
 $ hormone4       : num  4.2 6.8 3.2 2.7 7.2 1.6 9.1 4.5 5.2 8.3 ...
 $ hormone5       : num  322.7 190.9 153.7 225.7 99.9 ...
 $ hormone6       : num  166.1 41 140.8 93.4 24.7 ...
 $ hormone7       : num  5.7 6.8 5.4 12.7 1.6 9.9 7.5 4 4.2 1.7 ...
 $ hormone8       : num  129 260 81.1 148 121 120 135 199 149 238 ...
 $ hormone9       : num  NA NA NA NA NA NA NA NA NA NA ...
 $ hormone10_generated: num  0.922 0.166 0.405 0.441 0.199 ...
 $ hormone11      : num  NA NA NA NA NA NA NA NA NA 17 ...
 $ hormone12      : num  NA NA NA NA NA NA NA NA NA 11.3 ...
 $ hormone13      : num  NA NA NA NA NA NA NA NA NA 306 ...
 $ hormone14      : num  NA 52.6 74.2 NA NA NA NA NA NA ...
 $ lipids1        : num  0.4 0.54 0.45 0.38 0.48 0.34 0.49 0.62 1.1 0.51 ...
 $ lipids2        : num  5.05 3.9 3.5 4.84 3.96 3.85 3.93 4.3 3.89 3.72 ...
 $ lipids3        : num  1.23 1.67 1.29 1.32 1.85 1.42 1.26 1.3 1.11 0.78 ...
 $ lipids4        : num  3.64 1.98 2.01 3.35 1.89 2.28 2.45 2.72 2.28 NA ...
 $ lipids5        : num  0.18 0.25 0.2 0.17 0.22 0.15 0.22 0.28 0.5 NA ...
 $ carb_metabolism : num  5.41 4.5 5.57 5.95 4.15 4.66 4.44 5.32 4.71 4.39 ...
 $ lipid_pero1    : num  NA 2.82 2.16 2.44 1.98 2.36 3.06 2.48 1.83 3.62 ...
 $ lipid_pero2    : num  NA 1.36 0.72 1.86 0.86 1.17 1.64 1.22 1.35 1.26 ...
 $ lipid_pero3    : num  NA 0.71 0.27 1.46 0.33 0.62 0.76 0.81 0.65 0.43 ...
 $ lipid_pero4    : num  NA 1.51 0.64 1.32 2.04 2.61 1.77 0.58 0.39 2.02 ...
 $ lipid_pero5    : num  10.4 13.7 10.6 10.4 15.1 ...
 $ antioxidant1   : num  1.64 1.33 1.85 1.34 1.72 NA 1.7 1.31 1.58 NA ...
 $ antioxidant2   : num  2.24 2.48 2.19 1.58 2.14 2.14 2.22 2.02 2.15 2.36 ...
 $ antioxidant3   : num  1.91 1.98 1.92 1.86 2.34 1.83 2.08 1.98 2.24 2.11 ...
 $ antioxidant4   : num  5.21 5.06 9.58 7.46 9.22 7.62 13 7.8 8.85 8.4 ...
 $ antioxidant5   : num  0.38 0.39 0.49 0.43 0.86 0.38 0.94 0.47 1.12 0.74 ...
```

Missing Data



```
> sum(is.na(MD_df))           # Total NAs in entire dataset
[1] 6608

> colSums(is.na(MD_df))      # NA counts per column
```

record_id	hormone1	hormone2	hormone3
0	14	16	19
hormone4	hormone5	hormone6	hormone7
17	18	16	16
hormone8	hormone9	hormone10_generated	hormone11
19	1009	393	987
hormone12	hormone13	hormone14	lipids1
987	954	931	17
lipids2	lipids3	lipids4	lipids5
17	17	276	276
carb_metabolism	lipid_pero1	lipid_pero2	lipid_pero3
17	42	39	117
lipid_pero4	lipid_pero5	antioxidant1	antioxidant2
45	41	127	49
antioxidant3	antioxidant4	antioxidant5	
50	42	40	

Missing Data



```
> na_stats <- colMeans(is.na(MD_df)) * 100 # % missing data
```

```
> na_stats
```

record_id	hormone1	hormone2	hormone3
0.000000	1.219512	1.393728	1.655052
hormone4	hormone5	hormone6	hormone7
1.480836	1.567944	1.393728	1.393728
hormone8	hormone9	hormone10_generated	hormone11
1.655052	87.891986	34.233449	85.975610
hormone12	hormone13	hormone14	lipids1
85.975610	83.101045	81.097561	1.480836
lipids2	lipids3	lipids4	lipids5
1.480836	1.480836	24.041812	24.041812
carb_metabolism	lipid_pero1	lipid_pero2	lipid_pero3
1.480836	3.658537	3.397213	10.191638
lipid_pero4	lipid_pero5	antioxidant1	antioxidant2
3.919861	3.571429	11.062718	4.268293
antioxidant3	antioxidant4	antioxidant5	
4.355401	3.658537	3.484321	

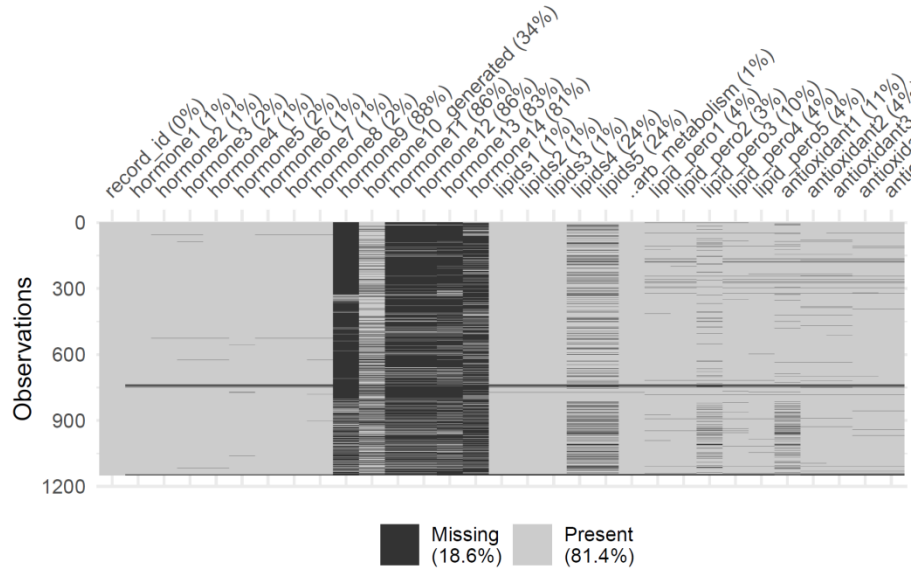
```
na_stats_filtered <- na_stats[na_stats <= 35] # missing data <=35 %
```

	Column	NA_Percent
1	record_id	0.000000
2	hormone1	1.219512
3	hormone2	1.393728
4	hormone3	1.655052
5	hormone4	1.480836
6	hormone5	1.567944
7	hormone6	1.393728
8	hormone7	1.393728
9	hormone8	1.655052
10	hormone10_generated	34.233449
11	lipids1	1.480836
12	lipids2	1.480836
13	lipids3	1.480836
14	lipids4	24.041812
15	lipids5	24.041812
16	carb_metabolism	1.480836
17	lipid_pero1	3.658537
18	lipid_pero2	3.397213
19	lipid_pero3	10.191638
20	lipid_pero4	3.919861
21	lipid_pero5	3.571429
22	antioxidant1	11.062718
23	antioxidant2	4.268293
24	antioxidant3	4.355401
25	antioxidant4	3.658537
26	antioxidant5	3.484321

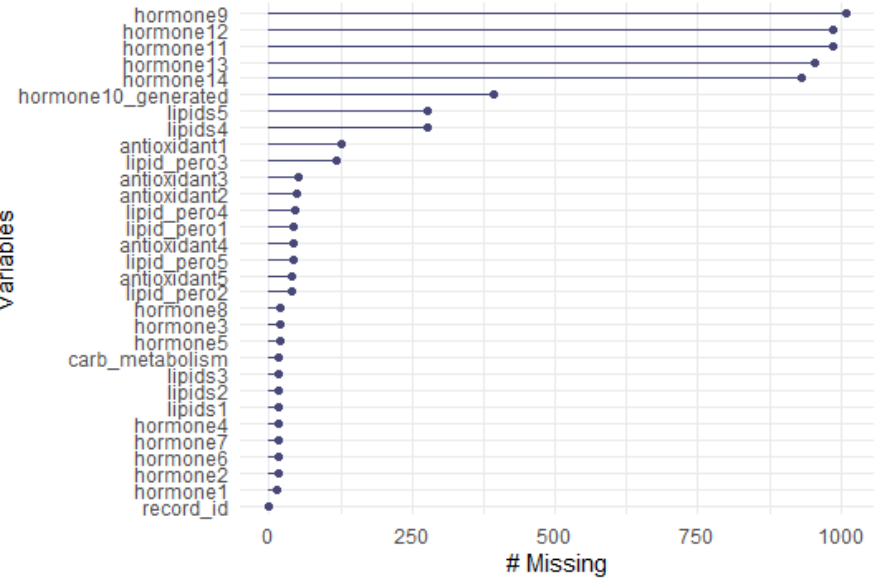
Missing Data



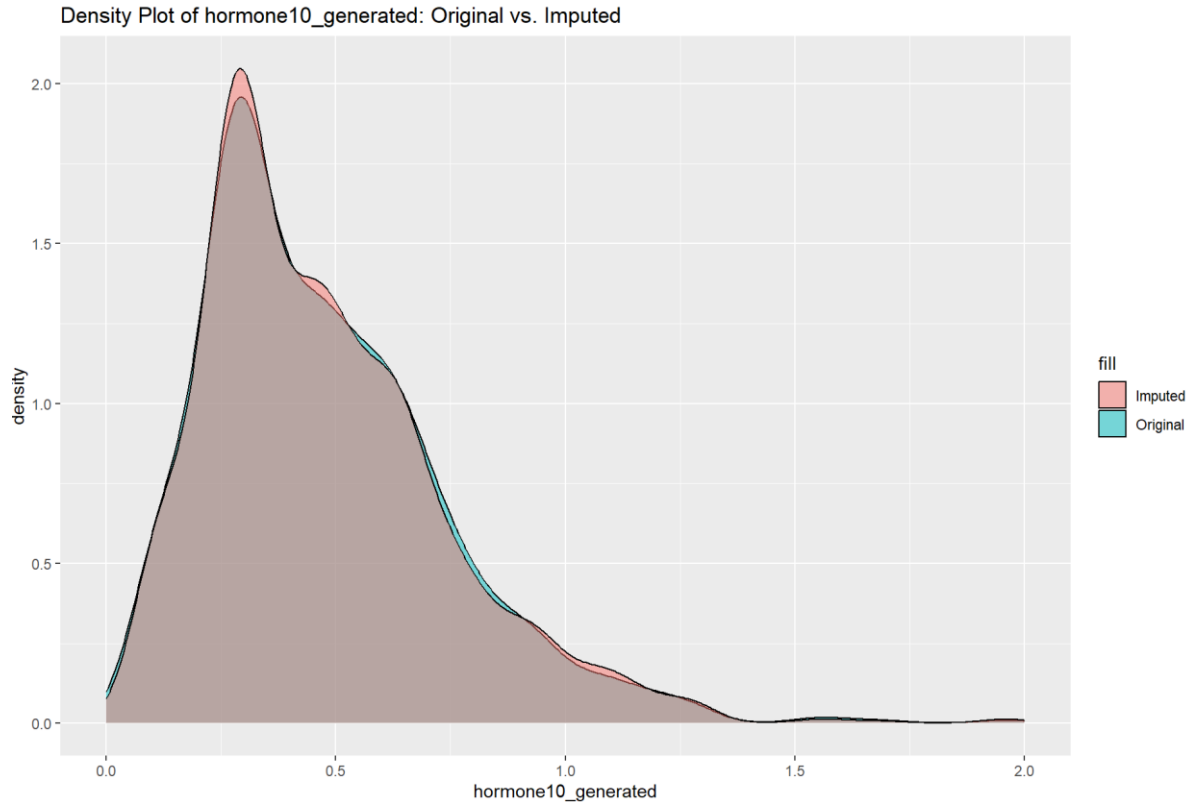
```
library(visdat)  
vis_miss(MD_df)
```



```
library(naniar)  
gg_miss_var(MD_df)
```



Missing Data



Missing Data

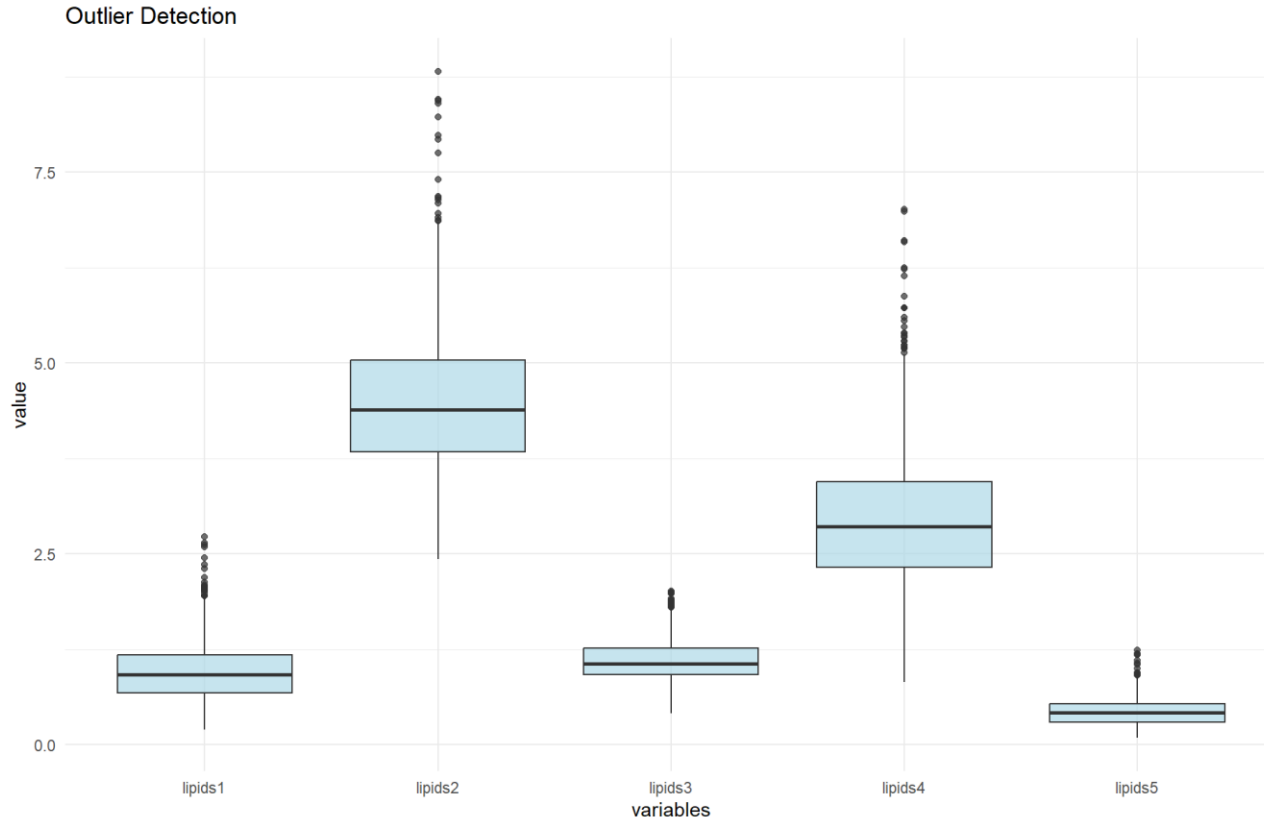


R-package **naniar**

Missing values, plotted with the `vis_miss` function

R-package **mice** (Multivariate Imputation via Chained Equations) - uses chained equations starting with the least missing

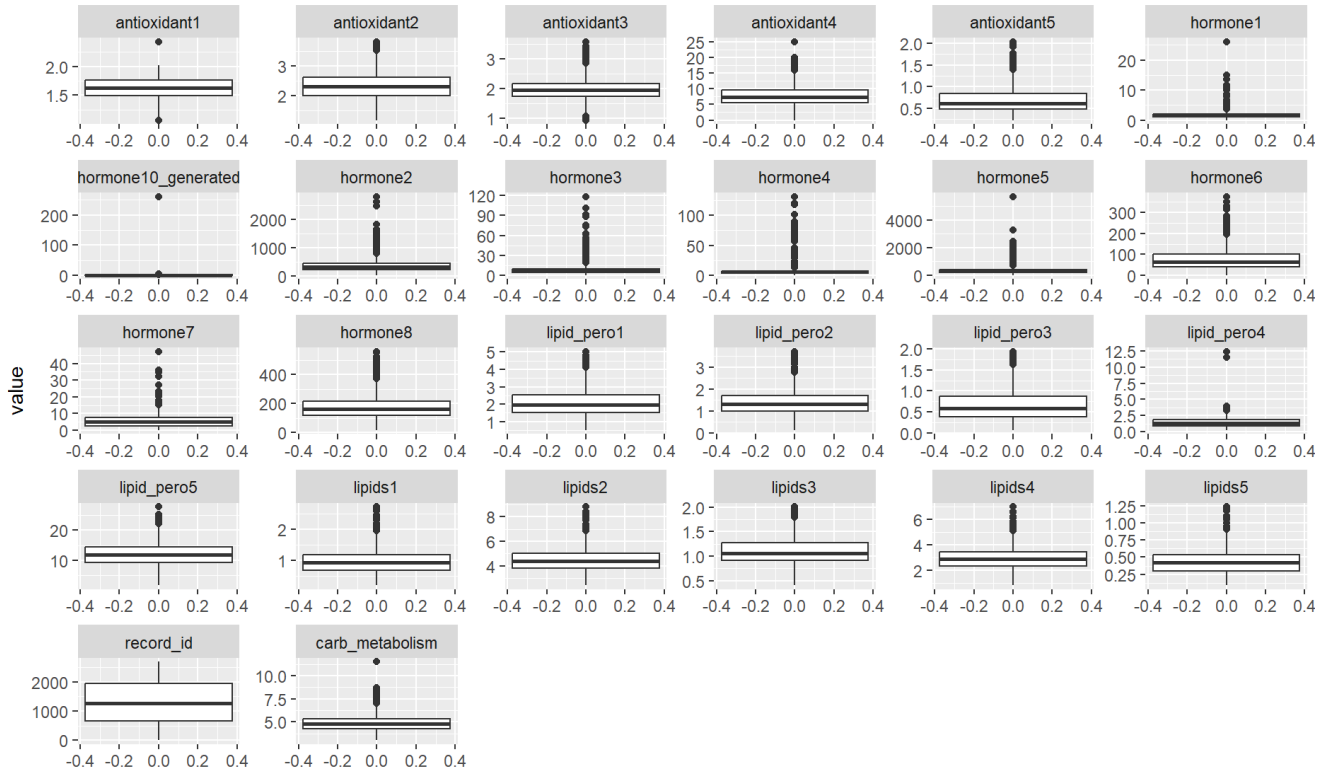
Outliers



Outliers



Boxplots for Outlier Detection



Task 1 (Deadline 30.03.2026 23:59)



Key Outputs to Check:

•p-value:

- $p > 0.05$ → Fail to reject H_0 (data is MCAR).
- $p \leq 0.05$ → Reject H_0 (data is not MCAR, likely MAR/MNAR).

Dataset “handle_MD_df” Perform Little's MCAR Test

1. For R 3.6.0-3.6.3

```
install.packages("BaylorEdPsych") # For Little's test
install.packages("mice") # For missing data visualization
library(BaylorEdPsych)
library(mice)
```

```
# Assuming your data is named 'df'
mcar_test <- LittleMCAR(df)
# View results
print(mcar_test)
```

2. For R 4.0+

```
library(naniar)
library(mice)
```

Task 1 (Deadline 30.03.2026 23:59)



- use default '**pmm**' method and compare with original, with '**rf**' method
- ***make a conclusion***

Imputation Method (method = 'pmm'):

PMM means Predictive Mean Matching and it's a non-parametric approach particularly suited for continuous data. PMM operates by finding observed values with similar predictive characteristics to the missing entries. The missing values are then imputed, thus preserving the distribution and variance of the original data more effectively than simpler methods, such as mean imputation.

Task 1 (Deadline 30.03.2026 23:59)



for extra points!

For dataset “imputed_handle_MD_df_final”
packages("dbscan")

Local Outlier Factor (LOF) algorithm for detecting outliers in multivariate data.

Calculating LOF factors

Visualizing results (Histogram of LOF factors, Bivariate scatterplot with outliers)